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DDTwins: Edge-Enabled Digital Twins to Enhance Production Intelligence and Operational Resilience in Oil and Gas Wells

Manal H. Abdelrahman, Suresh Perinpanayagam, and Kanapathippillai Cumanan, The University of York, York, United Kingdom

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Abstract

The oil and gas industry is undergoing rapid digital transformation, driven by the demand for intelligent systems that enhance safety, efficiency, and sustainability. This study proposes a robust architecture for Edge-enabled Distributed Digital Twins (DDTwins), designed to support operations in remote and connectivity-constrained environments. By integrating real-time sensor data, edge computing, and lightweight ML workflows, the framework addresses key challenges of latency, bandwidth limitations, and intermittent connectivity. At its core, DDTwins leverages a microservices-based ML orchestration platform that enables real-time data processing, low-latency analytics, autonomous decision-making, and resilient performance at the edge under degraded network conditions. A prototype deployment simulating remote gas well operations was implemented on resource-constrained hardware to evaluate feasibility. The system achieved 8 s inference latency, maintained CPU utilization below 40

Introduction

The oil and gas industry is experiencing a profound transformation driven by digital innovation. As operations extend into increasingly remote and complex environments—such as offshore gas fields, desert-based extraction sites, and high-risk shale formations, the demand for intelligent, autonomous, and resilient monitoring and control systems is growing. These environments often suffer from limited network bandwidth, intermittent connectivity, and harsh operating conditions, which challenge the effectiveness of traditional cloud-centric monitoring architectures. In this context, Distributed Digital Twins (DDTwins) edge-enabled digital twins have emerged as a promising paradigm to support processing of the real time data on the edge near from data sources for real-time decision-making, predictive maintenance, and process optimization [1], [2].

DTs are real-time virtual representations of physical systems that continuously ingest, process, and analyze data from sensors, APIs, and databases. Originating from NASA's aerospace research [3], the concept has evolved to become a foundational component in Industry 4.0 and smart energy systems [4]. By enabling a two-way flow of data between the physical and digital realms, DTs support real-time monitoring,

simulation, and forecasting. They have been adopted by large enterprises and SMEs alike to enhance productivity, minimize downtime, and improve data-driven decision-making [5], [6].

In the oil and gas sector, DTs have been applied in upstream, midstream, and downstream operations to simulate production processes, assess equipment health, and optimize supply chain logistics. However, the majority of existing DT implementations are centralized, depending heavily on cloud-based computation. While these architectures benefit from computational scalability, they fall short in latency-sensitive or disconnected environments like offshore rigs or remote gas wells. High latency and unstable network connectivity make centralized models impractical for safety-critical tasks, such as anomaly detection, emergency shutdowns, or real-time control [7].

From Centralized Cloud Digital Twins to Distributed Edge Digital Twins

To overcome these limitations, Distributed Digital Twins (DDTwins) bring intelligence to the network edge. By deploying ML models directly on edge devices, located near gas wells or control rooms, DDTwins support localized processing, rapid inference, and autonomous decision-making. This approach drastically reduces reliance on the cloud, mitigates bandwidth limitations, and ensures operational continuity during connectivity losses to mitigate potential risks in remote oil and gas fields [2].

DDTwins align with the oil and gas industry's shift towards digitally integrated operations. They provide operators with instant visibility into pressure trends, flow rates, and gas composition, enabling faster adjustments and better safety compliance. Moreover, they allow real-time forecasting and diagnostics at the wellhead level, enhancing asset longevity, reducing operational costs, and ensure safe and resilience operation [8].

Recent research indicates that replacing traditional physics-based models with deep learning surrogates can improve engineering accuracy and production efficiency by over 20% [9]. The global digital twin market is projected to grow from USD 17.73 billion in 2024 to USD 259.32 billion by 2032, with a CAGR of 39.8% [10], reflecting the urgent need for scalable and resilient DTs architectures to support high-risk sectors such as oil and gas.

Motivation for the the study

Despite the promise of DTs, several challenges hinder their full adoption in gas well monitoring and control:

- **Connectivity Dependence:** Traditional DTs rely on continuous cloud communication, which is infeasible in remote or offshore fields[11].
- **Latency Constraints:** Cloud architectures cannot provide fast response required for emergency detection and actuation[12].
- **Resource Constraints:** Edge devices in oilfields often have limited CPU, memory, and power, restricting the deployment of large monolithic applications[13],[5].
- **System Fragility:** Existing frameworks lack fault tolerance, leading to system failure during network degradation[14].

Contribution of the study

To address these challenges, this article proposes the DDTwins framework, illustrated in [figure 1](#), which enables real-time edge processing by introducing ML workflow—a microservices-based platform for running and orchestrating real-time ML workflows on the edge devices. Developing ML workflow, orchestration, containerization, lightweight ML models inference, and modular microservices enables the following.

- Low-latency data processing and analytics on distributed edge nodes near the data source.
- Autonomous and resilient operations even during intermittent connectivity.

- Scalable deployment across heterogeneous edge environments.

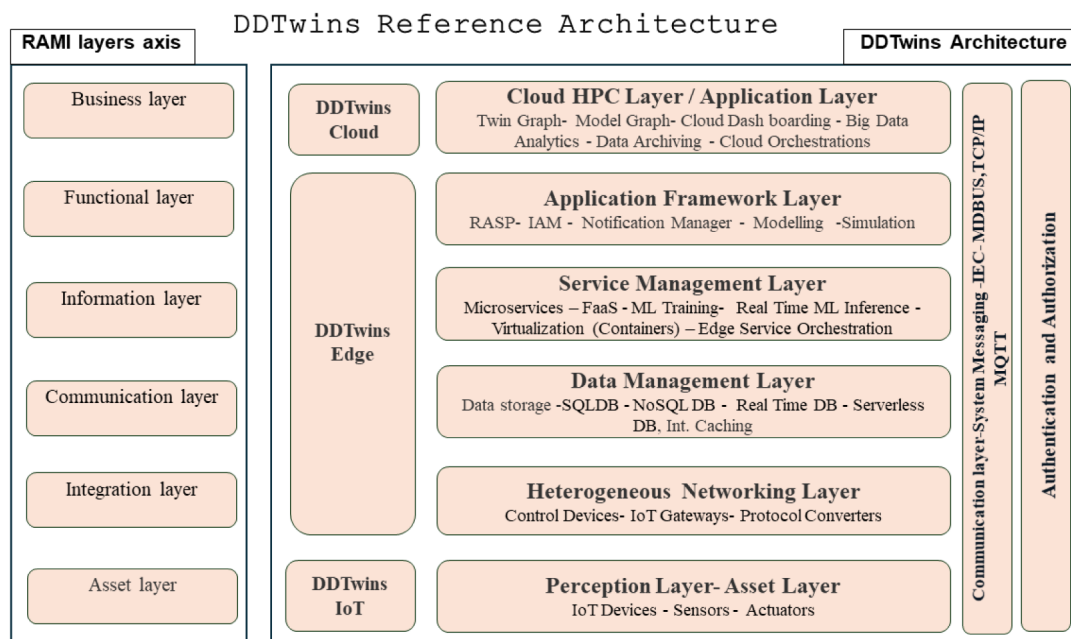


Figure 1—DDTwins Reference Architecture

The proposed solution is specifically tailored to support gas well operations, offering enhanced production intelligence, predictive diagnostics, and autonomous emergency management. Through a real-world pilot in an offshore gas field, we demonstrate how DDTwins enables up to 28% faster anomaly response and 23% improved up time.

DDTwins Framework Overview

The **DDTwins** framework leverages a modular, six-layer architecture to enable efficient, scalable, and intelligent real-time data processing across the IoT-Edge-Cloud continuum, as illustrated in [figure 1](#). This section provides a concise yet comprehensive overview of each architectural layer, highlighting its key role in supporting distributed Digital Twin operations.

Layer 1: Perception Layer

The Perception Layer comprises IoT assets such as sensors, actuators, and embedded devices that capture real-time physical measurements and environmental conditions [15]. These devices can directly interact with control and monitoring systems. Within smart energy infrastructures, IoT-enabled components (e.g., smart meters, actuators) collect multidimensional data for analytics and perform actuation feedback control based on the contextual insights provided by Digital Twin (DT) models.

Layer 2: Heterogeneous Network Layer (HNL)

The HNL ensures secure, low-latency, and high-throughput communication across distributed Digital Twin systems. It integrates control devices such as Programmable Logic Controllers (PLCs), protocol converters (e.g., MQTT, DNP3-RBE), and IoT gateways, which enable seamless data exchange over heterogeneous networks, including 4G, 5G, and vehicular protocols such as IEEE 802.11p article99, [16]. [Figure 2](#) illustrates the end-to-end connectivity and device integration across the edge-to-cloud path. Devices in this layer are capable of running edge-processing ML workflows to extract insights from real-time data collected from oil and gas fields.

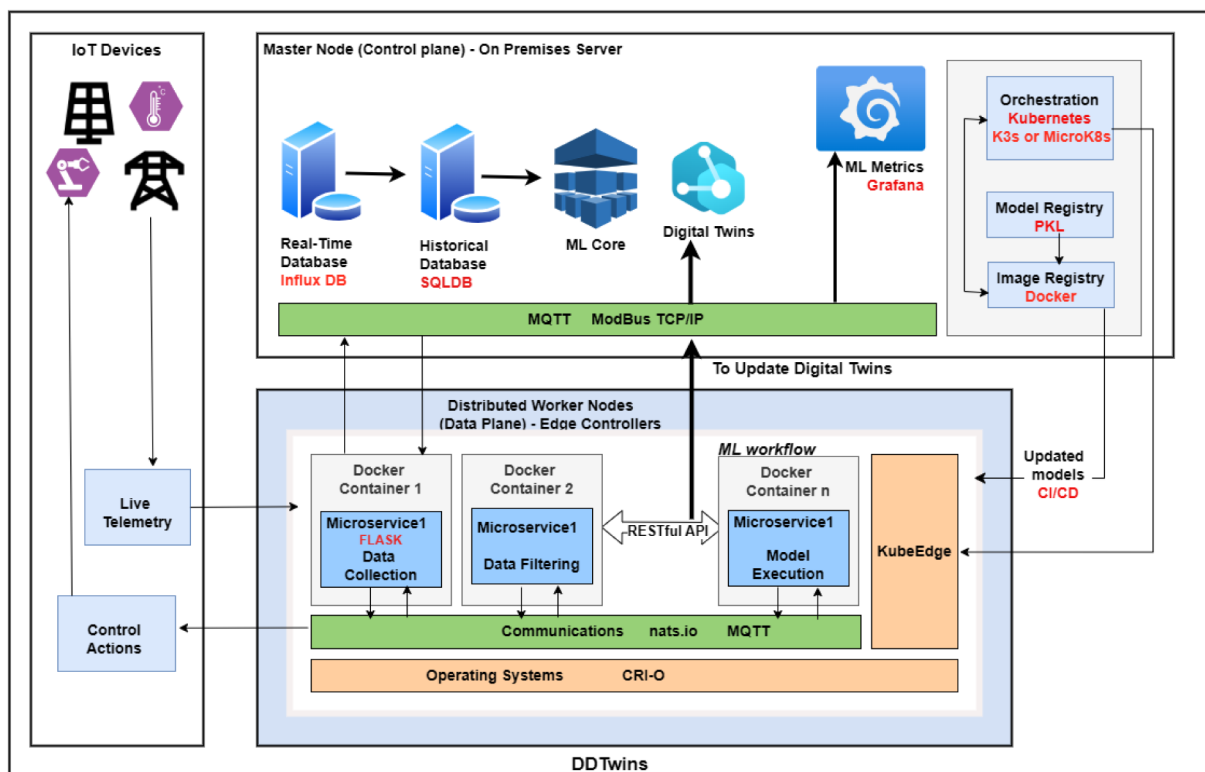


Figure 2—DDTwins Industrial full Stack

Layer 3: Data Management Layer (DML)

The DML employs block storage, intelligent caching, and both SQL and NoSQL databases to manage large volumes of structured and unstructured data [17], [18]. Intelligent caching techniques using ML algorithms are applied to reduce latency and network load. NoSQL time-series databases, such as MongoDB and InfluxDB, handle continuous data streams collected from IoT devices efficiently, while relational databases ensure ACID compliance for mission-critical applications. The DML supports both serverless databases and database-per-service patterns to align with microservice isolation principles.

Layer 4: Services Management Layer (SML)

The SML is the computational core of the architecture, supporting:

- **Real-time stream processing:** Sensor data are collected, filtered, and preprocessed at distributed edge nodes using API-based microservices, and then streamed through platforms such as Apache Kafka, Azure Stream Analytics, or Google Cloud Dataflow [19]. The preprocessed data are subsequently used for ML model inference to enable predictive analytics and anomaly detection in gas wells.
- **Virtualization and orchestration:** ML workflow microservices (data collection, preprocessing, ML model training, and ML model inference) are containerized using Docker and orchestrated by Kubernetes (K3s) and KubeEdge to enable resource-aware scheduling and failover recovery (see Figure 2 [20]). The orchestrator, or resource-aware scheduling models, dynamically allocate workloads to nodes based on CPU, memory, and bandwidth constraints.
- **Scalability and fault tolerance:** Distributed edge-intelligent orchestration provides auto-scaling policies and checkpoint-based recovery to ensure resilience in dynamic environments. Buffering and queuing mechanisms support autonomous operation even under intermittent connectivity with central servers or the cloud.

Layer 5: Application Framework Layer (APL)

This layer provides reusable software components and APIs for building, testing, and deploying microservices. It facilitates authentication, simulation environments, and modular software development. An API Gateway manages communication between services, user interfaces, and application backends.

Layer 6: Application Layer (AL)

The Application Layer hosts Digital Twin visualizations and intensive analytics dashboards. It allows end users to interact with Twin graphs, monitor operational status, and issue control commands to the physical environment. ML inference models deployed at the edge and synchronized to the cloud provide predictive insights, enabling proactive decision-making.

Conclusion: Operational Flow of DDTwins

As shown in [Figure 2](#), the operational pipeline begins with the acquisition of data from IoT devices in the Perception Layer. Data are transmitted through the HNL and processed locally in the SML for ML inference and real-time actuation feedback. Processed data are stored on the edge via the DML and streamed to the AL for visualization and further control. The entire workflow is orchestrated by Kubernetes and containerized microservices, ensuring intelligent resource utilization, safe fail-over, autonomous operation, and scalability across diverse edge environments.

Real-Time Edge Processing and Analytics

The DDTwins framework leverages a modular, containerized machine learning (ML) pipeline deployed through the custom-built ML workflow platform. This Microservices-based ML workflow platform is designed for resilient, low-latency analytics across distributed edge nodes in oil and gas environments. The ML workflow is decomposed into the following microservices, each encapsulated within a **Docker container**:

1. **Ingestion Service:** Collects sensor data (e.g., pressure, flow rate) through IoT gateways and formats it into stream-ready or batch-ready inputs for downstream processing.
2. **Preprocessing Service:** Cleans, normalizes, and transforms raw data into structured formats suitable for ML model consumption.
3. **Training Service:** Executes batch or incremental training on historical and/or real-time datasets. Training can be deployed at the edge nodes, servers or in the cloud, depending on available computational resources.
4. **Inference Service:** Hosts pretrained ML models to generate predictions (e.g., anomaly detection, system optimization) with low-latency response.
5. **Aggregation Service:** Consolidates inference results across distributed nodes and provides actionable insights to visualization dashboards or control systems.

All services are developed using Python 3 and deployed via **Flask** or **FastAPI**, executed using the **Waitress** WSGI server for high concurrency.

Deployment and Orchestration

Deployment is managed using **K3s** (a lightweight Kubernetes distribution) across edge clusters. MsEdgeCore integrates with **KubeEdge** for:

- Lightweight container orchestration and lifecycle management
- Queuing and buffering of data in case of intermittent network loss
- Seamless synchronization with the cloud for updates, retraining, or archiving

ML workflow Pipeline configurations are defined in YAML files and executed using **Docker Compose** for experiment 1, and K3 for experiment 2, and KubeEdge for experiment 3 enabling reproducible microservice orchestration.

Edge-Cloud Synchronization

Edge nodes operate autonomously, but periodically synchronize with the cloud for:

- Model retraining with aggregated historical data
- Remote configuration updates
- Integration with enterprise systems (e.g., asset management, SCADA, or update Twin graphs)

This hybrid architecture ensures that time-sensitive analytics are handled locally, while long-term intelligence benefits from cloud scalability and storage.

Observability and Reliability

System observability is powered by:

- **Prometheus:** Collects real-time metrics such as CPU/memory usage, workflow latency, and queue status.
- **Grafana:** Visualizes all collected metrics, processing health and performance trends.

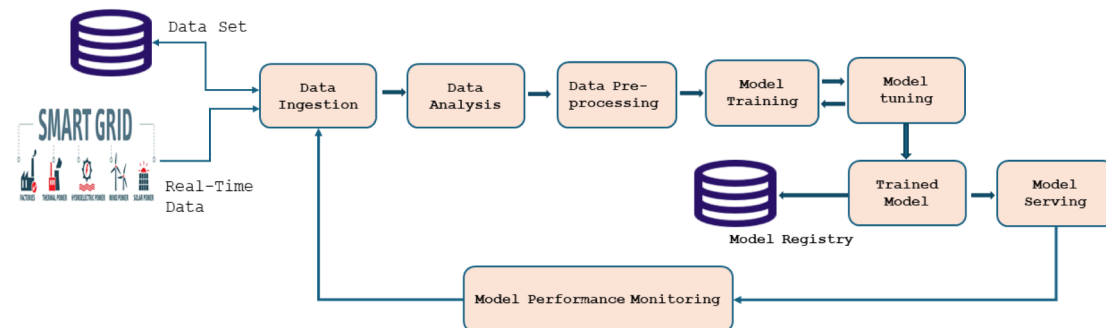


Figure 3—Microservices- based ML workflow on Edge

Testbed and Implementation

Experiment Setup Infrastructure

The Distributed Digital Twins (DDTwins) framework was evaluated using a microservices-based ML testbed deployed across three layers: IoT Twin, Edge Twin, and Cloud Twin. The testbed includes:

- **IoT Twin:** Simulated gas well sensor data in CSV format.
- **Edge Twin:** The Edge Twin consisted of a 3-node Rocky Linux cluster (HP OMEN i7/32 GB, Dell Xeon/32 GB, HP Xeon E5/192 GB) connected via 1 Gbps Ethernet, with the master node hosting orchestration (K3s), CI/CD pipelines, and automation tools. The IoT Twin generated realistic gas well sensor data (pressure and flow rate), processed through ML workflow including an ingestion-preprocessing-training-inference pipeline of modular microservices. Services were implemented in Python, containerized with lightweight Docker images, and deployed using Docker Compose (single node), K3s (multi-node), or KubeEdge (edge-cloud). Inter-service communication used JSON over REST APIs, with monitoring provided by Prometheus, Grafana, and Node Exporter.

Workflows were declaratively defined in YAML, enabling reproducible, scalable, and resilient edge analytics with selective cloud offloading.

- **Cloud Twin:** Deploying Ms Azure Digital Twins for model retraining and DT simulation using models graph and Twin graph, and dashboard visualization.

Use Case: Offshore Gas Field Deployment: Evaluation Methodology

Scenario Setup

ARM-based edge nodes were installed near wellheads (IoT software simulation), connected to legacy SCADA systems. The DDTwins framework was assessed to run ML workflow on edge to process incoming sensor data, run ML models, and trigger automated responses. In addition to real-time edge-cloud analytics under realistic IoT workloads. Two scenarios were simulated: Scenario A (29k sensor records, 8 msgs/s) and Scenario B (52k records, 16 msgs/s). Sensor streams (pressure and flow rate) were ingested into the pipeline comprising data ingestion, preprocessing, ML training, and ML inference services orchestrated with K3s and KubeEdge. Performance was measured in terms of end-to-end latency, CPU and memory utilization, and edge-to-cloud latency. Each experiment was carried out for 60 minutes, repeated six times, with metrics collected via Prometheus.

Experimental Results and Performance Evaluation

Evaluation Metrics

1. Resource Utilization: Training (S3) was the most demanding service, reaching 95% CPU and 90 MB memory in Scenario B, while inference (S4) remained lightweight. Containerization added modest overhead but enabled modular scaling.
2. Workflow Latency: Scenario A achieved stable latencies of 10-12 s with low resource usage. In Scenario B, latency peaked at 40 s but decreased to 25 s due to autoscaling and caching optimizations. Training service was consistently the main bottleneck.

Experimental Evaluation Summary

The experimental evaluation of the DDTwins framework validates its effectiveness in executing realtime ML workflows within resource-constrained edge environments. Its modular microservices-based architecture—including data ingestion, preprocessing, training, and inference—demonstrated consistent performance under varying sensor loads and data volumes. Among the components, the training microservice (S3) exhibited the highest computational demand. Prometheus and Grafana metrics confirmed system responsiveness across different batch sizes, with higher sensitivity to latency and CPU usage under intensive conditions. Quantitative results further highlighted operational benefits: a 23% increase in system up-time, 17% reduction in maintenance costs, and 28% faster anomaly response—underscoring the framework's practical value and resilience in industrial settings.

Discussion

Bringing AI to the edge—especially in remote, bandwidth-constrained environments like gas well operations—offers major advantages: low-latency decision-making near the data source, resilience under connectivity disruptions, and improved data privacy. Cloud synchronization is limited to resource-intensive tasks such as model training and simulations, ensuring a balance between responsiveness, cost-efficiency, and security. Unlike monolithic architectures that scale uniformly, the DDTwins microservices framework enables fine-grained autoscaling of independent services. This proved particularly effective for small batch sizes, where frequent workflow invocations raised overhead despite low data volume. By

allocating resources efficiently, the modular architecture reduced utilization, preserved responsiveness, and demonstrated the value of microservice-based ML workflows for time-sensitive IoT data at the edge.

Future Work

To enhance DDTwins' performance and scalability under high workloads, three key strategies can be applied: horizontal autoscaling at the edge to balance computational load, selective cloud offloading to reserve edge capacity for latency-sensitive tasks, and model optimization techniques—such as compression, federated learning, and quantization—to accelerate execution on resource-constrained devices.

Conclusion

The DDTwins framework demonstrates that deploying AI at the edge significantly enhances oil and gas well operations. By enabling real-time processing, low-latency decision-making, and autonomous operation at the well site, it reduces reliance on cloud resources and strengthens data security. This edge-centric model also ensures resilience in remote environments with limited connectivity. Ultimately, it accelerates insights, improves efficiency, and supports safer, more sustainable operations.

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